

Executive Summary

Obesity is a major public health concern in the United States and is associated with increased risk of chronic diseases and healthcare costs. The goal of this project was to identify key factors associated with obesity and develop a predictive model using nationally representative health data.

Data

- Source: National Health and Nutrition Examination Survey (NHANES)
- Survey Cycles: 2017–2020 and 2021–2023
- Final Dataset: 15,025 participants
- Features: Demographic characteristics and body measurements
- Target Variable: Obesity Status (BMI ≥ 30)

Methodology

- Merged demographic (DEMO) and body measurement (BMX) datasets using participant ID (SEQN)
- Cleaned and preprocessed data
- Imputed missing values and encoded categorical variables
- Developed a Logistic Regression model
- Applied backward stepwise elimination for feature selection
- Evaluated performance using train/test data and confusion matrices

Key Findings

- Body measurements were the strongest predictors of obesity.
- Waist circumference, hip circumference, and arm circumference showed the highest predictive value.
- Demographic variables contributed relatively little to model performance.
- The final model achieved:
 - Training Accuracy: 93.6%
 - Test Accuracy: 94.2%
 - Pseudo $R^2 > 0.75$

Business & Public Health Impact

- Supports early identification of individuals at higher risk of obesity.
- Provides insights for targeted prevention and intervention programs.
- Can assist public health organizations with resource planning and health education initiatives.
- Establishes a foundation for future dashboard development, model comparison, and deployment in decision-support tools.